Matrix Polynomial

Given $A \in \mathbf{R}^{n \times n}$ and $p(A)$ where the coefficients are given in $a = (a_0, a_1, \ldots, a_m)$ with

$$p(A) = a_0 I + a_1 A + \cdots + a_m A^m.$$

1. Find an eigenvector x_1 that corresponds to the eigenvalue $\lambda_k(p(A))$ that has the minimum modulus.
2. Find an eigenvector x_2 that corresponds to the eigenvalue $\lambda_k(p(A))$ that has the maximum real part.

Implementation

Goal: Fill the given function, then submit your function in Grader:

```
import numpy as np

def extract_eigenvec(A, a)::
    ...
    return x1,x2

if __name__ == "__main__":
    # write your test section here
    np.random.seed(67)
    n = 1000
    m = 50
    A = ...
    a = ...

    x1, x2 = extract_eigenvec(A, a)

    print(f"x1: {x1}")
    print(f"x2: {x2}")
```

Input Constraints

Input: All arrays are integers $\in [-5, 5]$

- $A \in \mathbf{R}^{n \times n}$ with entries in $(-1, 1)$ and $50 \leq n \leq 1000$
- $a \in \mathbf{R}^m$ with integer values and $4 \leq m \leq 50$

Sample Output

Note: The following outputs come from the test section.

```
x1:
 [ 4.61335566e-03+0.j -1.07329922e-02+0.j  1.52246339e-02+0.j
  ...
  5.77334838e-02+0.j -3.94747115e-02+0.j -3.14874051e-02+0.j
 -9.45981312e-04+0.j]
x2:
 [-8.06023706e-03+1.58265359e-02j  6.38410703e-03-2.01185264e-02j
  ...
 -1.58335385e-02-1.24954232e-02j -1.39319600e-02-9.16837080e-03j]
```